

Notes: Boustanifar and Kang (RFS, 2025)

The Brand Premium

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1 Big picture

- **Question.** Do firms with valuable brands earn abnormal stock returns, or are strong brands already fully priced by the market?
- **Core idea.** The paper argues that the usual *input* measure of brand value – cumulative advertising expenses – is a poor proxy for true brand strength. Instead, it uses two *output* measures of brand value:
 - Interbrand’s yearly *Best Global Brands* list, and
 - a new text-based measure built from how often firms discuss “brand” in annual filings.
- **Main finding.** A portfolio of top brands earns economically large and statistically significant abnormal returns. For the Interbrand-based portfolio, the abnormal return is about **22–30 basis points per month**, or roughly **3% per year**.
- **Mechanism.** The evidence points to a **mispricing** channel, not a standard risk-factor channel:
 - the premium is concentrated in firms that develop brands *internally*,
 - analysts systematically *underestimate* future earnings of top-brand firms,
 - top-brand firms earn higher abnormal returns around earnings announcements,
 - the premium is not explained by standard intangible-related factors or by overlap with “best workplace” firms.
- **Negative result.** When brand value is measured using the traditional input measure based on advertising expenses, the abnormal return disappears once profitability and investment factors are controlled for.
- **Economic takeaway.** Brand names are very visible, so one might expect them to be overpriced. The paper argues the opposite: *because accounting does a poor job recording internally created brand capital, and because the long-run payoff of strong brands is hard to forecast, valuable brands can be systematically undervalued.*

2 Data and measurement (key objects)

2.1 Why the paper rejects the standard input measure

- The standard finance literature often measures brand capital with the perpetual-inventory stock of advertising expenses.
- The paper argues this measure is problematic for at least five reasons:
 - around **70%** of Compustat firms do not report advertising expense separately,
 - many top brands are therefore missing from the input-based measure,
 - advertising is an *input*, not an output, and says little about investment quality,
 - it is backward-looking and misses sudden rises or declines in brand value,
 - some highly valuable brands spend little or no traditional advertising.

- The authors emphasize that advertising is only one component of brand building. Customer service, product reputation, platform effects, and brand loyalty can all matter even if reported advertising is low.
- **Interpretation.** The paper’s central empirical move is to replace a noisy, incomplete, backward-looking measure with output-based, forward-looking measures of brand importance.

2.2 Interbrand Best Brands (BB) data

- The main output measure is the annual **Interbrand Best Global Brands** list.
- Interbrand has published the list since **2000**. The first list had 75 brands, and from 2001 onward the list contains 100 brands.
- Interbrand’s valuation method has three main steps:
 1. estimate the economic profit attributable to the brand,
 2. compute the *Role of Brand Index* (RBI), measuring how much purchases are due to the brand rather than other product attributes,
 3. discount brand earnings using a measure of *brand strength*.
- To be included, a brand must be global, visible, have public financial data, and be expected to generate positive long-term economic profits.
- The paper maps brands to *U.S. public stocks* because returns come from CRSP. Since one company can own multiple top brands, the BB list is converted from a brand list to a stock list.
- The portfolio is rebalanced on the **first day of the month after** the Interbrand announcement. This timing lets the market react to the public announcement first and then asks whether the market still underprices future brand-related cash flows.
- The Interbrand-based return sample runs from **September 2000 to December 2020**, for **244 months**.

2.3 Text-based output measure of brand value

- The second output measure is built from annual SEC filings.
- The authors download annual filings from EDGAR beginning in **1996**, remove duplicates and amendments, and count the frequency of the word *brand* (and *brands*) in each filing.
- The normalized text-based brand measure is

$$TB_{it} = \frac{\text{brand-count}_{it}}{\text{total words}_{it}}.$$

- After linking filings to CRSP/Compustat, the data contain **113,007 firm-year observations** and **12,854 unique firms**.
- The text-based portfolio includes firms that:
 - rank among the **top 100** or **top 200** firms by normalized brand frequency,

- have market capitalization of at least **\$5 billion**,
- have December fiscal year-end.
- The portfolios are rebalanced at the end of the following June.
- The correlation between the equal-weighted top-100 text portfolio and the Interbrand BB portfolio is about **85%**.
- The paper gives intuitive examples:
 - in 2020, a typical firm mentions “brand” about **3** times,
 - Nike mentions it **210** times,
 - Marriott **119** times,
 - Coca-Cola **96** times.
- This measure also captures firms such as Tesla, Alphabet, Twitter, Uber, Salesforce, LinkedIn, and Grubhub – firms that may not look like top brands under the advertising-based measure.

2.4 Portfolio construction and descriptive facts

- The average yearly number of BB stocks in the Interbrand portfolio is about **47**.
- The list is persistent but not fixed. On average, **3** firms are added and **2** are dropped each year.
- Of firms on the original 2000 BB list, about **29%** had dropped out by 2010 and **45%** by 2020.
- BB firms appear in **27 of 48** Fama–French industries. The most common industries include:
 - computer software,
 - computers,
 - business services,
 - consumer goods,
 - restaurants and hotels,
 - banking,
 - electronic equipment,
 - machinery.
- The industry mix evolves over time. Software and business services become more prominent later in the sample, while consumer goods, restaurants/hotels, and banking are persistently represented.
- BB firms are very large:
 - in **2020**, mean market cap is about **\$288 billion**, median about **\$143 billion**, and max about **\$2.3 trillion**,
 - about **90%** of BB firms are above the NYSE 80th size percentile.
- This matters because it shows the results are *not* driven by microcaps.
- Table 3 also shows that reported intangible assets among BB firms average roughly:

- **13.6%** of assets in 2000,
- **21.0%** of assets in 2020.
- But these reported intangibles are mainly *acquired* intangibles, not internally developed brand capital.

2.5 Input measure of brand capital

- For comparison with prior literature, the paper constructs a traditional input measure of brand capital using the perpetual inventory method.
- Using advertising expenses AD_{it} , the stock evolves as

$$BC_{it} = (1 - \delta)BC_{i,t-1} + AD_{it},$$

with depreciation rate

$$\delta = 0.20.$$

- To initialize the stock, the paper assumes advertising growth of **10%** per year.
- The sample for the input-based measure starts in **1975**, because before that very few firms reported advertising expenses.
- **Interpretation.** This input-based measure is used mainly as a falsification/comparison exercise: if the brand premium is really about output brand strength rather than ad spending, then the input measure should perform worse.

3 Models and empirical specifications (all equations + interpretation)

3.1 Factor regressions: measuring abnormal returns

The baseline time-series regression is

$$ER_t = \alpha + \sum_k \beta_k \text{Factor}_{k,t} + \varepsilon_t,$$

where ER_t is the portfolio excess return over the risk-free rate.

- α is the abnormal return not explained by factor exposure.
- The paper uses five asset-pricing models:
 - Carhart four-factor,
 - Fama–French 5-factor plus momentum (FF6),
 - q factor model,
 - $q5$ model,
 - AQR 6-factor model.

Interpretation. This is the paper’s first pass at asking whether top-brand firms earn abnormal returns after controlling for the standard factor zoo.

3.2 Main factor-regression results

- The equal-weighted Interbrand BB portfolio earns monthly abnormal returns of:
 - **0.30%** in Carhart,
 - **0.22%** in FF6,
 - **0.30%** in q ,
 - **0.25%** in $q5$,
 - **0.25%** in AQR.
- The average estimated alpha across these models is about **26 bps per month**, i.e. roughly **3% per year**.
- The point is not that one specific model gives the strongest result. The point is that the premium survives across many modern factor models.

3.3 Why weighting matters: equal-weighted vs value-weighted

- The equal-weighted BB portfolio shows a robust premium.
- A straight value-weighted portfolio produces much weaker, often insignificant, alphas.
- The paper shows this is *not* because the equal-weighted result is driven by tiny firms. Rather, the straight value-weighted portfolio is dominated by a few gigantic firms.
- To address this, the authors also use **capped value-weighted** portfolios, capping weights at the NYSE 90th or 80th percentile.
- With capped weights, the premium returns:
 - FF6 alpha about **27 bps** with 90th-percentile cap,
 - FF6 alpha about **32 bps** with 80th-percentile cap.
- **Interpretation.** The brand premium is not a microcap artifact. It is strongest in a balanced portfolio, but it is washed out when a few mega-cap names dominate all weights.

3.4 Fama–MacBeth predictive regressions

The predictive cross-sectional specification is

$$r_{i,t+1} = \alpha_{0t} + \alpha_{1t}BB_{it} + \alpha_{2t}Z_{it} + \varepsilon_{it},$$

where:

- $r_{i,t+1}$ is next-month stock return,
- BB_{it} indicates whether stock i is on the most recent Best Brands list,
- Z_{it} contains characteristics such as beta, size, book-to-market, momentum, asset growth, profitability, illiquidity, R&D, idiosyncratic volatility, skewness, and, in augmented regressions, advertising, brand capital, and organization capital.

Interpretation. This regression asks whether BB stocks earn higher future returns than otherwise similar firms, controlling directly for firm characteristics.

3.5 Fama–MacBeth results

- In the baseline Fama–MacBeth regression, the BB coefficient is about **0.38%** per month.
- In the fuller specification with many controls, the BB coefficient is about **0.32%** per month.
- When the regressions also include advertising, brand capital, and organization capital, the BB premium remains significant and is often even larger.
- **Interpretation.** The premium is not simply a disguised size, value, momentum, profitability, or investment effect. Even after conditioning on a large set of characteristics, BB stocks still predictably outperform.

3.6 Time variation in returns

- The paper checks whether the average premium is driven by only a few extreme months.
- It is not. Estimated monthly excess returns remain positive in all major subperiods:
 - dot-com crash,
 - 2002–2008 boom,
 - the 2008–2009 financial crisis,
 - post-crisis / sovereign-debt period,
 - 2014–2020.
- The corresponding monthly excess returns are about **43, 11, 158, 37, and 30** basis points.
- In five-year blocks, the premium is also positive in each block:
 - 2000–2005: **31** bps,
 - 2006–2010: **43** bps,
 - 2011–2015: **11** bps,
 - 2016–2020: **41** bps.
- The cumulative excess return in the Fama–MacBeth setup reaches about **196%** over the full sample.
- **Interpretation.** The premium is persistent, not a one-off historical accident.

3.7 Best brands versus other intangible-related explanations

The paper explicitly asks whether the BB premium is really something else in disguise.

- **Best workplaces / employee satisfaction.** The paper controls for the return to the “Best Companies to Work For” portfolio and for whether firms are in that list. The BB alpha barely changes:
 - FF6 alpha falls only from **22** bps to **20** bps,
 - Fama–MacBeth BB coefficient falls only from **0.32** to **0.29**.

- **Organization capital and other intangible factors.** Augmenting FF6 with an organization-capital factor or with input-based brand-capital portfolios does not explain the BB premium.
- **General takeaway.** The premium is not explained by the broader “intangibles” literature alone. The paper argues there is a distinct *brand* effect.

3.8 Internally generated versus externally acquired intangibles

This is one of the most important mechanism tests in the paper.

- Accounting rules expense *internally generated* brand investment but allow *acquired* identifiable intangibles to be capitalized.
- Therefore, if mispricing comes from brands being missing from accounting statements, the premium should be stronger for firms that build brands internally.
- The paper splits BB firms into:
 - **HIGH INT BB:** firms with high reported intangible assets,
 - **LOW INT BB:** firms with lower reported intangibles.
- The logic is that high reported intangibles proxy for more externally acquired intangibles; low reported intangibles proxy for more internally generated brand capital.
- The results show:
 - HIGH INT BB coefficients are small and often insignificant,
 - LOW INT BB coefficients are much larger, typically around **0.38–0.46** in the Fama–MacBeth specifications.
- **Interpretation.** This sharply supports the accounting-mispricing story. Once brand value is more likely to be already recorded on the balance sheet, the premium becomes much smaller.

3.9 Earnings surprise mechanism

To test whether the market underestimates the future earnings power of strong brands, the paper runs

$$\text{Earnings Surprise}_{it} = b_0 + b_1 BB_{it} + b_2 Z_{it} + \varepsilon_{it}.$$

- One- and two-year earnings surprises are actual EPS minus median analyst forecast, scaled by price.
- Long-term surprise is actual 5-year EPS growth minus median analyst long-term growth forecast.
- The BB coefficients are positive and significant:
 - 1-year surprise: about **2.19, 2.29, 1.80**,
 - 2-year surprise: about **3.74, 3.89, 2.09**,
 - 5-year surprise: about **3.70, 1.43, 1.31**.
- **Interpretation.** Analysts systematically underforecast the future earnings of best-brand firms.

3.10 Earnings announcement returns

The paper also studies announcement-window returns.

- The dependent variable is cumulative abnormal return over the $[-1, +1]$ window around the earnings announcement.
- In the full specification, BB firms earn an additional abnormal return of about **0.28%** around earnings announcements.
- Assuming four earnings announcements per year, the paper argues that this explains **more than 40%** of the FF6 alpha.
- **Interpretation.** This is exactly what one would expect under underreaction to future profitability: a significant part of the premium is realized when earnings news arrives.

3.11 Text-based output measure results

The text-based measure serves as a powerful out-of-sample test.

- For the **top-100** text-based portfolio:
 - equal-weighted FF6 alpha is about **0.57%** per month for 1996–,
 - equal-weighted FF6 alpha is about **0.71%** per month for 2000–,
 - capped value-weighted FF6 alpha is about **0.38%**.
- For the **top-200** text-based portfolio:
 - equal-weighted FF6 alpha is about **0.62%** per month for 1996–,
 - equal-weighted FF6 alpha is about **0.63%** per month for 2000–,
 - capped value-weighted FF6 alpha is about **0.25%**.
- These are often larger than the Interbrand-based results.
- **Interpretation.** The premium is not an artifact of Interbrand’s list or media coverage around its publication. A completely different output-based measure gives similar or even stronger results.

3.12 Input-measure results

The paper finally compares its preferred output measures with the classic input measure.

- For the top-100 long-only advertising-based brand-capital portfolio, the Carhart alpha is positive but the FF6 alpha drops to about **0.12%** and becomes statistically insignificant.
- In long-short input-based portfolios, the FF6 alpha can even become negative in the full sample, around **-0.39%**.
- The paper shows that any apparent input-based premium is explained by exposure to profitability and investment factors.
- **Interpretation.** The brand premium is about *valuable brands*, not about *high advertising stocks*.

4 Identification and instruments (what makes the estimates credible)

4.1 What is hard here

- This is not a clean causal design. The paper explicitly acknowledges that it does *not* have a natural experiment that randomly assigns brand value across firms.
- So the main question is not “Is the estimate causal?” but rather: *Does the paper make a compelling mispricing case using multiple pieces of evidence?*

4.2 What identifies the paper’s empirical claims

- **Measurement upgrade.** The first key move is replacing the advertising-based input measure with forward-looking output measures that better capture actual brand strength.
- **Layered asset-pricing controls.** The premium survives many factor models and Fama–MacBeth characteristics.
- **Not small stocks.** The result survives excluding small firms and in capped value-weighted portfolios.
- **Not generic intangibles.** The premium survives controls for brand capital, organization capital, expected growth, and workplace-quality effects.
- **Mechanism evidence.** The premium is much stronger for internally generated brands than for externally acquired intangibles, which is exactly what the accounting-information channel predicts.
- **Forecast-error evidence.** Analysts underpredict brand firms’ future earnings, and announcement returns around earnings releases are elevated.

4.3 Why the output measures help

- Interbrand is explicitly designed to value the *economic output* of the brand, not merely brand-building expenses.
- The text-based measure captures brand salience and strategic emphasis even when advertising is missing or close to zero.
- Because both measures lead to similar conclusions, the interpretation is much more persuasive than if only one measure worked.

4.4 What remains a limitation

- The results remain subject to noncausal interpretations.
- There may still be omitted variables correlated with brand value and future returns.
- Interbrand is a curated list and could contain media-attention effects, though the text-based measure reduces this concern.

- The paper makes the strongest possible case available in this setting, but it does *not* claim clean causal identification.

5 Tables and figures

5.1 Table 1 and Table 2; Figures 1–3: what are the top-brand firms?

Table 1
Summary statistics on number of BBs used in the paper

Year of list	# of BB stocks	# added	(%)	# dropped	(%)
2000	38				
2001	51	13	(25.5)	0	(0.0)
2002	53	6	(11.3)	4	(7.8)
2003	51	0	(0.0)	2	(3.8)
2004	49	2	(4.1)	4	(7.8)
2005	45	3	(6.7)	7	(14.3)
2006	45	1	(2.2)	1	(2.2)
2007	47	3	(6.4)	1	(2.2)
2008	46	4	(8.7)	5	(10.6)
2009	45	4	(8.9)	5	(10.9)
2010	46	3	(6.5)	2	(4.4)
2011	45	1	(2.2)	2	(4.3)
2012	48	3	(6.3)	0	(0.0)
2013	48	3	(6.3)	3	(6.3)
2014	48	1	(2.1)	1	(2.1)
2015	48	1	(2.1)	1	(2.1)
2016	48	2	(4.2)	2	(4.2)
2017	47	2	(4.3)	3	(6.3)
2018	46	0	(0.0)	1	(2.1)
2019	48	2	(4.2)	0	(0.0)
2020	46	2	(4.3)	4	(8.3)
Average	47	3	(5.8)	2	(5.0)

This table reports (in column 2) the number of BB stocks per year that are included in the BB portfolio, that is, those that had returns available on CRSP for at least 1 month during the portfolio formation periods of each year. Column 3 shows the number of companies that are added to the list, and column 4 reports the number of added companies as a fraction of total number of companies in the list. Columns 5 and 6 repeat the similar statistics (the raw number and the proportion) for companies that are dropped from the list in each year.

Table 2
Best brands, by industry

Industry	Brands
Computer Software (115)	Amazon.com (21), Microsoft/LinkedIn (21), Oracle (19), Google/YouTube (16), Yahoo! (13), Adobe (12), Facebook/Instagram (9), Salesforce (4)
Computers (91)	Apple (21), HP/Compaq (21), Xerox (16), Dell (14), IBM (14), Sun Microsystems (3), Compaq (2)
Business Services (84)	Accenture (19), eBay (17), Visa (13), MasterCard (9), IBM (7), AOL/Time (5), Hewlett Packard Enterprise (5), PayPal (5), Disney (4)
Consumer Goods (68)	Colgate (21), Harley-Davidson (19), P&G/Gillette/Pampers/Duracell (18), Kodak (8), Estee Lauder (1), Xerox (1)
Restaurant, Hotels, Motels (66)	McDonald's (21), Starbucks (21), KFC/Pizza Hut (20), Hilton (2), Burger King (1), Marriott International (1)
Banking (61)	American Express (21), Citigroup (21), JPMorgan (19)
Electronic Equipment (52)	Cisco (21), Intel (21), Motorola (9), Texas Instruments (1)
Machinery (50)	GE (21), Caterpillar (19), Deere & Company (10)
Trading (47)	Goldman Sachs (20), Morgan Stanley (19), Merrill Lynch (8)
Food Products (43)	Kellogg's (21), Heinz (13), Kraft (7), Campbell's (2)
Candy & Soda (42)	Coca-Cola/Sprite (21), Pepsi (21)
Apparel (33)	Nike (21), Polo Ralph Lauren (12)
Communication (32)	MTV (17), Discovery (7), AT&T (3), Disney (3), PayPal (1), Zoom (1)
Pharmaceutical Products (29)	Johnson & Johnson (19), Pfizer (6), Merck (4)
Transportation (29)	UPS (16), FedEx (11), UBER (2)
Automobiles and Trucks (27)	Ford (21), Chevrolet (3), Tesla (3)
Beer & Liquor (26)	Jack Daniel's (15), Corona (11)
Entertainment (18)	Disney (14), Netflix (4)
Construction Materials (17)	3M (12), Gillette/Duracell (5)
Business Supplies (16)	Kleenex (16)
Retail (15)	GAP (15)
Tobacco Products (11)	Marlboro (11)
Recreation (4)	Barbie (4)
Aircraft (4)	Boeing (4)
Petroleum and Natural Gas (4)	ExxonMobil (4)
Insurance (3)	AIG (2), Berkshire Hathaway (Heinz) (1)
Personal Services (1)	Hertz (1)

This table reports the brand names counts in each industry group. The numbers in the industry column indicate the total count of stock-years for the best brands in that industry. The numbers in the Brands column show how many years each company was listed as part of the best brands. In cases in which one firm has multiple brands, such as Microsoft/LinkedIn, Google/YouTube, Facebook/Instagram, HP/Compaq, AOL/Time, Gillette/Pampers/Duracell, KFC/Pizza Hut, and Coca-Cola/Sprite, we count the brand as one stock.

- **Read.** These tables and figures establish the sample and show that BB firms are large, diversified across industries, and evolve gradually through time.
- Table 1 shows that the stock list is persistent but not fixed. This matters because the paper is not simply buying the same fixed basket forever.
- Table 2 and Figures 1–2 show that brand-intensive firms come from many industries, not just one sector.
- Table 3 and Figure 3 show BB firms are overwhelmingly large-cap. This is crucial because it defeats the “maybe this is just a small-stock anomaly” objection.

5.2 Table 4 and Table 5: baseline abnormal returns

- **Read.** Table 4 is the cleanest statement of the main result: top brands earn positive alphas under modern factor models.
- Table 5 shows that the result is strongest in equal-weighted and capped value-weighted portfolios, not because of microcaps but because straight value-weighting overconcentrates the portfolio in a few giants.

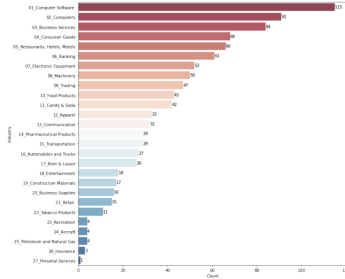


Figure 1
Counts of best brands' stocks, by industry
This figure shows total counts of Best Brands' stocks in each Fama-French 49 industry category during our sample, from September 2000 to December 2020.

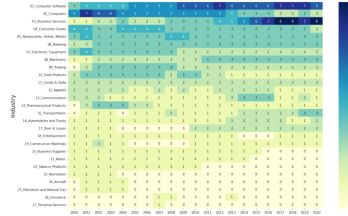


Figure 2
Evolution of best brands' industry affiliations
This figure shows the number of Best Brands' stocks by industry and year. The darker the cell color, the larger the number.

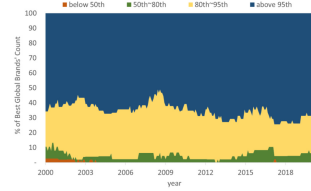


Figure 3
Size distribution of best Brands' stock across time
This figure plots the percentage of number of Best Brands' firms that belong to each NYSE size category for each year from 2000 to 2020. The categories are between 50th and 80th percentiles (large caps), between 80th and 95th percentiles (mega caps), and above 95th percentile (giga caps). NYSE different percentile breakpoints are taken from Kenneth French's website.

Table 4
Excess returns of BB portfolio

Dependent variable:	(1)	(2)	Excess return		
Model:	Carhart	FF6	q	q5	AQR
α	0.30*** (3.37)	0.22** (2.60)	0.30*** (3.18)	0.25** (2.11)	0.25** (2.59)
β_{MKT}	0.98*** (35.16)	1.01*** (37.36)	1.00*** (32.79)	1.04*** (25.37)	1.05*** (33.63)
β_{SMB}	-0.02 (-0.48)	0.01 (0.12)			-0.01 (-0.14)
β_{HML}	0.02 (0.45)	-0.05 (-1.12)			0.04 (0.74)
β_{MOM}	-0.10** (-2.55)	-0.11*** (-2.91)			
β_{RMW}		0.11* (1.68)			
β_{CMA}		0.11* (1.73)			
β_{R_ME}			-0.05 (-1.37)	-0.05 (-1.19)	
β_{R_IA}			0.14** (2.08)	0.02 (0.25)	
β_{R_ROE}			-0.14** (-1.99)	-0.30*** (-3.22)	
β_{R_EG}				0.20** (2.43)	
β_{UMD}					-0.09** (-2.38)
β_{BAB}					-0.06 (-1.64)
β_{QMJ}					0.23*** (3.63)
# obs	244	244	244	244	244

This table reports time-series factor regression results of BB equal-weighted portfolio. The dependent variable is the monthly return of BB portfolio in excess of the risk-free rate. We report the results for all five factor models (one per column). The factor models used are Carhart four-factor, Fama-French 5-factor plus momentum (FF6), q, q-5, and AQR 6-factor model. The factors in FF6 are market (Mkt), size (SMB), value (HML), momentum (MOM), profitability (RMW), and investment (CMA). The q-model has market, size (R_ME), investment (R_IA), and profitability (R_ROE), and the q5 model adds to these the expected investment growth factor (R_EG). The AQR model includes the market, size, value, and momentum (UMD), and adds to them betting against beta (BAB) and quality minus junk (QMJ) factors. Newey-West standard errors are calculated, and t -statistics are reported in parentheses. The sample is from September 2000 to December 2020. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 5
Sensitivity of excess returns of BB portfolio to weighting methods

Dependent variable:	(1)	(2)	Excess return		
Model:	Carhart	FF6	q	q5	AQR
α original estimates (ew)	0.30 (3.37)	0.22 (2.60)	0.30 (3.18)	0.25 (2.11)	0.25 (2.59)
α excluding small caps	0.30 (3.31)	0.22 (3.00)	0.30 (2.97)	0.24 (2.10)	0.25 (2.75)
α straight vw	0.12 (1.26)	0.14 (1.45)	0.16 (1.54)	0.02 (0.19)	0.11 (1.08)
α capped vw 90th	0.28 (3.17)	0.27 (3.04)	0.33 (2.99)	0.23 (2.00)	0.28 (2.93)
α capped vw 80th	0.34 (3.49)	0.32 (3.24)	0.38 (3.10)	0.28 (2.21)	0.34 (3.03)
# obs	244	244	244	244	244

This table reports the sensitivity of the excess return of equal-weighted BB portfolio with respect to excluding relatively smaller (below NYSE 50th percentile) BBs and value-weighting instead of equal-weighting. We provide excess return estimates for both the straight and capped (at the 90th and 80th NYSE size percentiles) value-weighted results. We report the excess return using five different factor models (one per column). The factor models used are Carhart four-factor, Fama-French 5-factor plus momentum (FF6), q, q-5, and AQR 6-factor model. The factors in FF6 are market (Mkt), size (SMB), value (HML), momentum (MOM), profitability (RMW), and investment (CMA). The q-model has market, size (R_ME), investment (R_IA), and profitability (R_ROE), and the q5 model adds to these the expected investment growth factor (R_EG). The AQR model includes the market, size, value, and momentum (UMD), and adds to them betting against beta (BAB) and quality minus junk (QMJ) factors. Newey-West standard errors are calculated, and t -statistics are reported in parentheses. The sample is from September 2000 to December 2020.

5.3 Table 6 and Figure 4: cross-sectional and time-series robustness

Table 6

Excess returns of the BB portfolio: Fama-MacBeth regressions

Dep var:	(1)	(2)	(3)	(4)	(5)
			Stock return ($t+1$)		
BB	0.38** (2.44)	0.32** (2.27)	0.31*** (2.73)	0.50*** (2.64)	0.52*** (2.87)
BETA	0.05 (0.27)	0.23 (1.36)	0.41** (2.14)	0.35** (2.28)	0.35** (2.29)
SIZE	-0.06 (-0.90)	-0.14*** (-3.43)	-0.10** (-2.40)	-0.12* (-1.95)	-0.13*** (-2.61)
BM	0.23 (1.51)	0.13 (0.97)	0.16 (1.42)	0.25** (2.52)	0.25** (2.05)
MOM	0.32* (1.69)	0.14 (0.71)	0.34** (2.40)	0.45** (2.36)	0.45** (2.37)
ASSET GROWTH		-0.55*** (-4.90)	-0.63*** (-4.91)	-0.69*** (-2.94)	-0.72*** (-3.49)
PROFIT		0.27* (1.77)	0.48*** (3.45)	0.67*** (2.86)	0.66*** (2.62)
ILLIQ		0.02*** (3.84)	0.02*** (3.13)	0.01 (1.08)	0.01 (0.98)
R&D		-0.11 (-0.69)	0.46* (1.67)	0.23 (0.55)	0.32 (0.87)
IDIO VOL		-0.20*** (-3.81)	-0.15* (-1.70)	-0.06 (-0.44)	-0.05 (-0.41)
SKEW		-0.09 (-1.32)	-0.02 (-0.35)	-0.02 (-0.27)	-0.02 (-0.27)
ADVERT			0.18 (0.16)	0.32 (0.26)	0.40 (0.32)
BRANDCAP				-0.12 (-0.45)	-0.07 (-0.25)
ORGCAP					-0.01 (-0.18)
Observations	886,440	857,032	362,914	117,469	117,469
R-squared	.04	.05	.06	.09	.09

This table presents the results (i.e., the average slopes) from predictive cross-sectional Fama-MacBeth regressions. The first specification uses characteristics associated with the Carhart model. The second model is considered the full model with a host of other characteristics for which we have nonmissing values for a large set of observations. The last three models add some extra control variables. Market beta (BETA) is estimated from a regression of monthly excess stock returns on monthly excess market return over a 60-month rolling window. Firm size (SIZE) is measured as the logarithm of the market capitalization and the book-to-market ratio (BM) is measured as of December of the previous year excluding negative book equity firms. Momentum (MOM) is measured as a stock's cumulative return over the past 12-month. We then add the two additional characteristics corresponding to factors of Fama-French 5-factor model together with some further control variables including: ASSET GROWTH is the growth in total assets over the most recent fiscal year, following Cooper, Gulen, and Schill (2008). Operating profitability (PROFIT) is computed as revenue minus cost of goods sold, selling, general and administrative expense, and interest and related expenses, deflated by book equity. Amihud illiquidity (ILLIQ) is computed as the absolute daily return divided by the daily dollar trading volume, averaged over all trading days in a month. R&D is research and developments expenses scaled by market capitalization. Idiosyncratic volatility (IDIO VOL) is defined as the standard deviation of the residuals from a regression of daily excess stock returns on the daily excess market return over a 1-month window. Skewness (SKEW) is the skewness of daily stock returns. ADVERT is advertising expenses scaled by market capitalization. BRANDCAP and ORGCAP are the stock of brand and organization capital (scaled by total assets) that are computed using the perpetual inventory method and the same parameters and sample selection as in Belo, Lin, and Vitorino (2014) and Eisfeldt and Papanikolaou (2013), respectively. All independent variables in the FM regressions are winsorized at the 1st and 99th percentiles. The total number of observations across all regressions and the average R-squared are reported. Newey-West standard errors are calculated, and t -statistics are reported in parentheses. The sample is from September 2000 to December 2020. * $p < .1$; ** $p < .05$; *** $p < .01$.

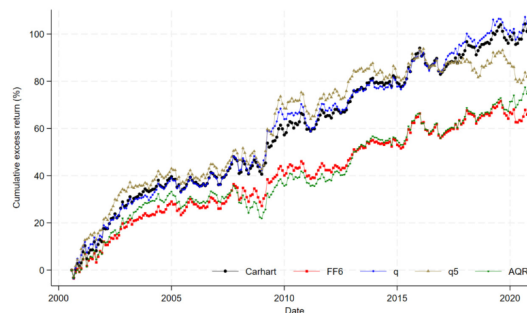


Figure 4

Cumulative excess returns of BBs

This figure plots the cumulative monthly excess returns (i.e., alpha plus residuals) of BB portfolio estimated using different factor models. The sample includes 244 months ranging from September 2000 to December 2020.

- **Read.** Table 6 shows the premium survives direct controls for many stock characteristics in Fama-MacBeth regressions.
- Figure 4 shows cumulative excess returns steadily trend upward rather than coming from a few lucky months.
- The five-year subsample evidence is especially useful for exam prep: the premium remains alive in each block of the sample.

5.4 Table 8: is this really just another intangible or workplace premium?

- **Read.** This is an important horse-race table.
- The BB premium is not explained by overlap with “Best Companies to Work For” or by common intangible-related factor stories.
- This strengthens the interpretation that *brand value specifically* matters.

5.5 Table 9: internally versus externally generated intangibles

- **Read.** This is the key mechanism table.

Table 8
Best brands versus best workplaces

	(1)	(2)	(3)	(4)
<i>A. Factor regressions</i>				
BB alpha	0.30*** (3.37)	0.27*** (3.16)	0.22*** (2.60)	0.20*** (2.35)
BC		0.13 (1.61)		0.13 (1.56)
Factor model	Carhart	Carhart	FF6	FF6
<i>B. Fama-MacBeth regressions</i>				
BB	0.38** (2.44)	0.34** (2.42)	0.32** (2.27)	0.29** (2.21)
BC		0.20* (1.94)		0.17* (1.96)
Controls	Baseline	Baseline	Full	Full

This table studies whether the BB excess return is driven by the correlation of Interbrand's Best Brands (BB) list and the list Best Companies to Work For in America (BC). The latter is a yearly list of best workplaces (or places with highest employee satisfaction), studied in Edmans (2011). The first panel our original results of Carhart and Fama-French 5-factor plus momentum (FF6) factor regressions (columns 1 and 3) together with the corresponding abnormal returns when we add the excess return to the BC portfolio as an additional right-hand-side variable (columns 2 and 4). Panel B report our original Fama-MacBeth estimates for the baseline (column 1) and full (column 3) models together with the corresponding results when we add an additional dummy variable indicating whether a company is part of the most recent BC list (columns 2 and 4). Newey-West standard errors are calculated, and *t*-statistics are reported in parentheses. The sample is from September 2000 to December 2020. **p* < .1; ***p* < .05; ****p* < .01.

Table 9
Reported intangibles and excess return

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
HIGH INT BB	0.24 (1.64)	0.20* (1.75)	0.22* (1.68)	0.17 (1.35)	0.10 (0.78)	0.06 (0.82)	0.09 (0.78)	0.04 (0.37)
LOW INT BB	0.40** (2.12)	0.43** (2.53)	0.38** (2.14)	0.38** (2.09)	0.39** (2.52)	0.46*** (3.64)	0.39*** (2.65)	0.38*** (2.60)
Main controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Advertising	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Brand/Org cap	No	No	No	Yes	No	No	No	Yes
Require nonmissing additional controls	–	Yes	No	No	–	Yes	No	No
Ind-adjusted returns	No	No	No	No	Yes	Yes	Yes	Yes
Observations	857,032	362,914	857,032	857,032	853,771	361,386	853,771	853,771

This table repeats the results of Fama-MacBeth regressions as in Table 6 but reports the excess returns separately for Best Brands (BB) that have relatively high intangibles (HIGH INT BB) versus those with relatively low intangibles (LOW INT BB) as reported in their balance sheet. HIGH INT BB is an indicator variable that is 1 if intangible assets reported in the balance sheet over total assets of a BB is above median in the year and 0 otherwise. The dependent variable in columns 1–4 is raw return while it is the industry-adjusted return in columns 5–8. Within each set, the first specification is our full specification corresponding to the second column in Table 6. The second specification (columns 2 and 6) reports results when we add advertising expenses. The third specification within each set (columns 3 and 7) is similar to the second specification with the exception that observations with missing advertising are replaced with 0. The last specification within each set (columns 4 and 8) follows a similar procedure as in the third specification with two additional control variables of brand capital and organization capital. All the control variables' definitions and details of the specifications are reported in the notes to Table 6. Newey-West standard errors are calculated, and *t*-statistics are reported in parentheses. The sample is from September 2000 to December 2020. **p* < .1; ***p* < .05; ****p* < .01.

- If accounting mismeasurement is the reason brand firms are undervalued, the premium should be concentrated in firms whose brand capital is *missing* from the balance sheet.
- That is exactly what the table shows: firms with lower reported acquired intangibles have much larger abnormal returns.

5.6 Table 10 and Table 11: analysts underreact to future brand earnings

Table 10
Earnings surprises

	(1)	(2)	(3)
<i>A. 1-year earnings</i>			
BB	2.19*** (3.96)	2.29*** (3.94)	1.80*** (2.59)
BM		0.10 (0.62)	0.11 (0.70)
SIZE			0.48 (1.62)
# obs	71,090	61,884	61,884
<i>B. 2-year earnings</i>			
BB	3.74*** (3.10)	3.89*** (3.12)	2.09*** (2.55)
BM		0.35 (1.29)	0.39 (1.46)
SIZE			1.82*** (3.25)
# obs	49,755	43,579	43,579
<i>C. Long-term earnings</i>			
BB	3.70*** (4.43)	1.43*** (2.69)	1.31*** (2.33)
BM		1.65*** (5.63)	1.63*** (5.57)
SIZE			2.12*** (3.19)
# obs	13,832	11,848	11,845

This table reports the results of regressions of earnings surprises on an indicator variable for whether the company was in the most recent Best Brand list (BB), and other control variables (book-to-market ratio and size). The dependent variable in panel A (B) is the actual EPS minus the I/B/E/S median analyst forecast 8 (20) months before the end of the forecast period, scaled by the stock price. In panel C, long-term earnings growth surprise is the actual five-year annualized EPS growth rate minus the median analyst long-term growth forecast. All coefficients are multiplied by 1000. All regressions include year and industry fixed effects as well as a constant. The sample is from September 2000 to December 2020. **p* < .1; ***p* < .05; ****p* < .01.

Table 11
Earnings announcement returns

	(1)	(2)	(3)	(4)
Dep var:		<i>CAR around earnings announcements</i>		
BB	0.25** (2.41)	0.25** (2.31)	0.28** (2.49)	0.28** (2.51)
BM				0.12*** (3.96)
SIZE				0.01 (0.90)
Year FE	No	Yes	Yes	Yes
Industry FE	No	No	Yes	Yes
Observations	258,621	258,621	258,621	253,110

This table shows the results of regressions of earnings announcement returns on an indicator variable for whether the company was in the most recent Best Brand (BB) list. The dependent variable is the cumulative abnormal return defined as the difference between the firm's compounded stock return and the value-weighted market return over the 3-day window $[-1, +1]$ around the date of the earnings announcement. Book-to-market ratio and size, both computed at the previous year-end, are used as control variables. The sample is from September 2000 to December 2020. **p* < .1; ***p* < .05; ****p* < .01.

- **Read.** These tables connect the asset-pricing result to a concrete forecast-error mechanism.
- Table 10 shows that analysts underestimate future earnings for brand firms over one-, two-, and five-year horizons.
- Table 11 shows that these mistakes are corrected around earnings announcements, where BB firms earn higher abnormal returns.

5.7 Table 12 and Table 13: output measure versus input measure

Table 12
Excess return to a text-based measure of brand value

	(1) Carhart	(2) FF6	(3) q	(4) q5	(5) AQR
<i>A. Top 100 firms</i>					
ew (1996–)	0.96*** (5.69)	0.57*** (4.02)	0.73*** (4.11)	0.58*** (3.36)	0.54*** (3.59)
ew (2000–)	1.12*** (6.68)	0.71*** (5.36)	0.99*** (6.32)	0.84*** (5.75)	0.71*** (5.03)
vw	0.58*** (3.33)	0.21 (1.40)	0.38** (2.21)	0.28* (1.67)	0.14 (0.89)
vw - capped	0.76*** (4.77)	0.38*** (3.02)	0.62*** (4.08)	0.50*** (3.52)	0.34** (2.57)
<i>B. Top 200 firms</i>					
ew (1996–)	0.97*** (5.49)	0.62*** (4.23)	0.76*** (4.26)	0.82*** (2.82)	0.63*** (3.80)
ew (2000–)	1.04*** (5.37)	0.63*** (4.43)	0.92*** (4.77)	0.96*** (3.06)	0.69*** (4.05)
vw	0.40*** (2.79)	0.12 (1.09)	0.24* (1.94)	0.21* (1.79)	0.13 (1.08)
vw - capped	0.60*** (4.53)	0.25*** (2.82)	0.46*** (3.85)	0.37*** (3.53)	0.27*** (2.74)

This table reports time-series factor regression results of BB portfolios constructed using textual analysis of companies' annual filings. The text-based measure of brand value is constructed using the count of words *brand(s)* in each filing divided into the total number of words in the document. To be included in the portfolio in a particular year, a company must meet the following two criteria: (a) ranked among top brands (i.e., top 100 in panel A and top 200 in panel B) in the year according to the text-based measure, and (b) has a market capitalization greater than or equal to \$5 bn. The dependent variable is the monthly return of BB portfolio in excess of the risk-free rate. We report alphas for five different factor regressions and for three different weighting schemes: equal-weighted (ew), value-weighted (vw), and capped value-weighted (vw-capped). We also report results (only ew for brevity) using the full sample starting from 1996. Newey-West standard errors are calculated, and *t*-statistics are reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 13
Brand capital and excess returns using the input measure

Sample:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Interbrand sample (2000–2020)				1975–2020			
Portfolio:	High Book brands		High - Low		High Book brands		High - Low	
Model:	Carhart	FF6	Carhart	FF6	Carhart	FF6	Carhart	FF6
α	0.30** (2.49)	0.12 (1.23)	0.26 (0.94)	−0.24 (−0.95)	0.26*** (3.08)	0.12 (1.40)	−0.04 (−0.19)	−0.39** (−2.03)
β_{MKT}	0.97*** (27.68)	1.03*** (39.59)	0.04 (0.63)	0.18*** (3.07)	1.03*** (35.63)	0.10*** (39.56)	0.14*** (2.99)	0.19*** (4.06)
β_{SMB}	0.26*** (4.20)	0.30*** (4.76)	−0.56*** (−3.11)	−0.37*** (−2.96)	0.10* (1.69)	0.17*** (3.97)	−0.80*** (−7.61)	−0.55*** (−6.95)
β_{HML}	0.12* (1.82)	−0.03 (−0.57)	0.58*** (5.03)	0.28*** (2.72)	0.20*** (3.20)	0.07 (1.31)	0.54*** (3.63)	0.34*** (3.63)
β_{MOM}	−0.21*** (−4.75)	−0.23*** (−5.72)	−0.03 (−0.35)	−0.11* (−1.97)	−0.19*** (−4.39)	−0.21*** (−5.46)	−0.03 (−0.31)	−0.09 (−1.21)
β_{RMW}		0.22*** (3.23)		0.82*** (6.95)		0.27*** (3.81)		0.85*** (9.64)
β_{CMA}		0.27*** (4.33)		0.22 (1.26)		0.23*** (3.81)		0.15 (0.97)
# obs	244	244	244	244	534	534	534	534

This table reports time-series factor regression results of equal-weighted portfolios sorted according to stock of firm level brand capital. Brand capital is measured using the perpetual inventory method, following [Belo, Lin, and Vitorino \(2014\)](#). The results are reported using two different portfolios: long-only top 100 brands in each year (columns 1, 2, 5, and 6), and a long-short portfolio which is long in stocks ranked on the top quintile of brand capital and short in the bottom quintile (columns 3, 4, 7, and 8). The table also reports the result corresponding to Interbrand sample period (columns 1–4) as well as going back to 1975, the year before which very few firms were reporting advertising expenses. The results for two factor models are reported: Carhart and Fama-French 5-factor plus momentum (FF6). The factors in FF6 are market (Mkt), size (SMB), value (HML), momentum (MOM), profitability (RMW), and investment (CMA). Newey-West standard errors are calculated, and *t*-statistics are reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

- **Read.** Table 12 is the strongest external validation of the paper: the text-based output measure gives similar or stronger abnormal returns.
- Table 13 provides the negative test: once brand value is measured using advertising-based stock of brand capital, the premium disappears after richer factor controls.
- Together, these two tables sharply separate *brand strength* from *advertising intensity*.

6 Short summary

- **Question.** Do top-brand firms earn excess returns?
- **Method.** The paper studies two output measures of brand value – Interbrand Best Brands and a text-based filing measure – and tests portfolio alphas with factor regressions and Fama–MacBeth regressions.
- **Main result.** Top-brand portfolios earn robust abnormal returns of about **3% per year** using Interbrand data, and even larger abnormal returns using the text-based measure.
- **Mechanism.** The premium is strongest for internally developed brands, analysts underforecast future earnings of brand firms, and brand firms earn positive abnormal returns around earnings announcements.
- **Negative result.** The traditional advertising-based input measure of brand capital does *not* generate robust abnormal returns once richer controls are added.
- **Interpretation.** The evidence is consistent with a **mispricing-based brand premium**.

7 Conclusion

7.1 Core conclusion

The paper concludes that output-based measures of brand value identify a substantial and robust *brand premium* in stock returns, while the standard input-based advertising measure does not.

7.2 Economic intuition

The intuition is straightforward. A valuable brand creates long-run cash-flow advantages through loyalty, pricing power, and resilience. But internally generated brand capital is largely invisible in accounting statements, because the investment is expensed rather than capitalized. As a result, the market and even professional analysts can understate the long-run earnings power of strong brands.

7.3 Takeaway

This paper is useful both substantively and methodologically. Substantively, it says that strong brands appear undervalued, not overvalued. Methodologically, it shows that *how* one measures an intangible matters enormously: output-based brand measures reveal a premium, while input-based advertising measures do not. For studying intangible assets in finance, the measurement lesson is as important as the return result itself.